

Project 2

Instagram User Analytics

(By Mohit Bohra – Data Analytics Trainee Trainity)

Project Description:- This project looks at how people use Instagram by running SQL queries on a database. The goal is to find useful information for the product team. like who the oldest and inactive users are, which hashtags are most popular, and whether any users might be fake. The main aim is to help Instagram grow by making smart decisions based on data.

Approach:

- Built the database and tables based on the given SQL structure.
- Developed SQL queries to answer each question outlined in the project description.
- Collected query results and observations using MySQL Workbench.
- Organized the results and insights into a well-structured report with screenshots.

Tech-Stack Used

- MySQL Workbench 8.0 , Ms Word and Gdrive

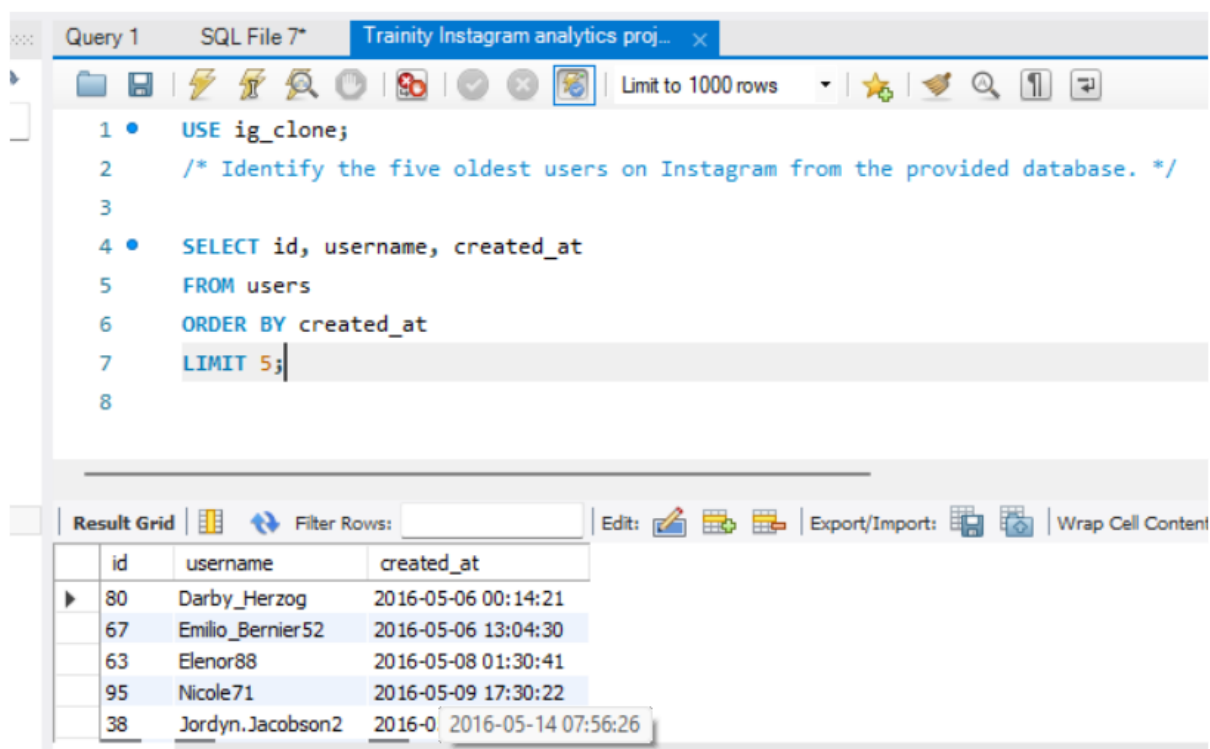
SQL Tasks :

A) Marketing Analysis:

1.Loyal User Reward:

The marketing team wants to reward the most loyal users, i.e., those who have been using the platform for the longest time.

Task: Identify the five oldest users on Instagram from the provided database.



The screenshot shows a SQL IDE window titled "Trainity Instagram analytics proj...". The query editor contains the following SQL code:

```
1 • USE ig_clone;
2   /* Identify the five oldest users on Instagram from the provided database. */
3
4 • SELECT id, username, created_at
5   FROM users
6  ORDER BY created_at
7  LIMIT 5;
```

Below the query editor, the "Result Grid" tab is active, displaying the results of the query. The results are as follows:

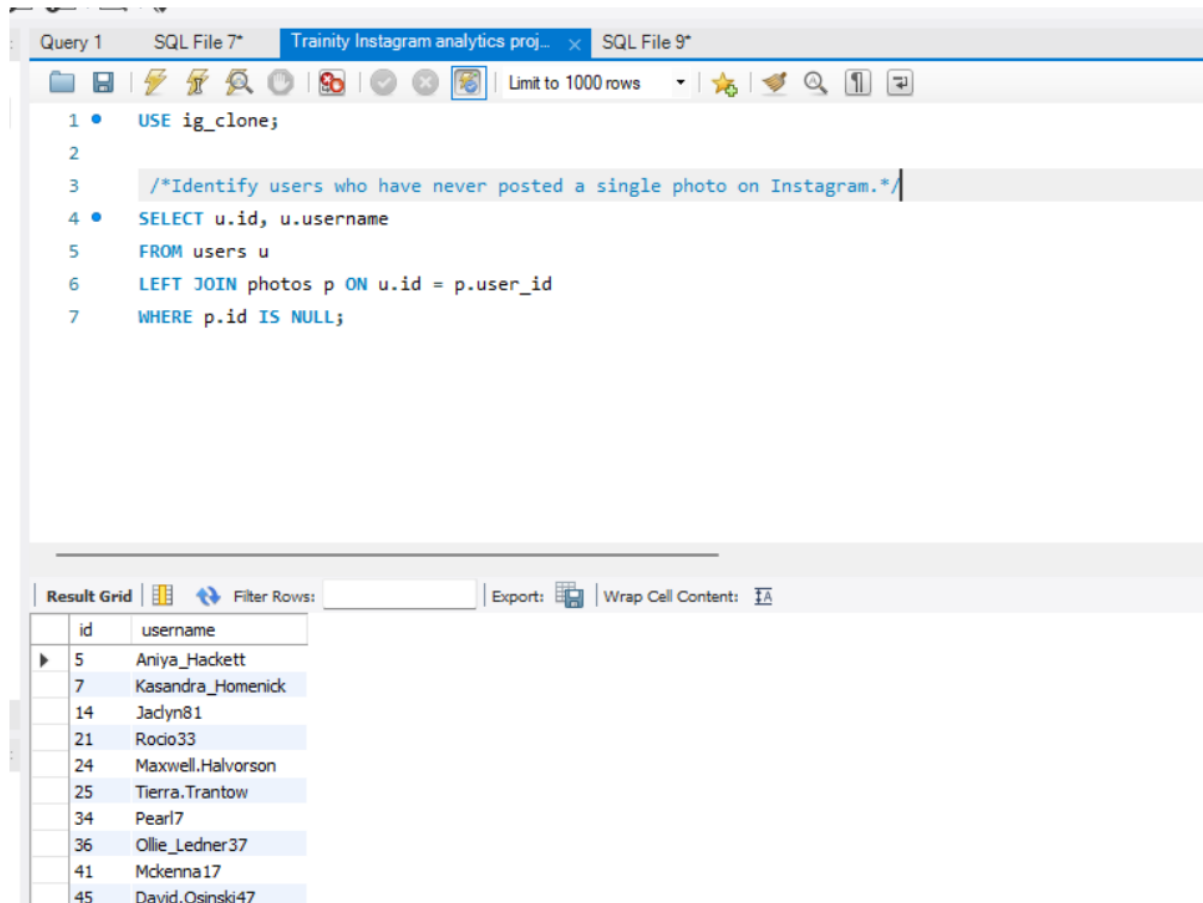
	id	username	created_at
▶	80	Darby_Herzog	2016-05-06 00:14:21
	67	Emilio_Bernier52	2016-05-06 13:04:30
	63	Elenor88	2016-05-08 01:30:41
	95	Nicole71	2016-05-09 17:30:22
	38	Jordyn.Jacobson2	2016-05-14 07:56:26

Insights: These users were the first to join the platform and have remained active the longest. They can be great choices for loyalty rewards

2. Inactive User Engagement:

The team wants to encourage inactive users to start posting by sending them promotional emails.

Task: Identify users who have never posted a single photo on Instagram.



The screenshot shows a SQL IDE interface with a query editor and a result grid. The query editor contains the following SQL code:

```
1 • USE ig_clone;  
2  
3 /*Identify users who have never posted a single photo on Instagram.*/  
4 • SELECT u.id, u.username  
5 FROM users u  
6 LEFT JOIN photos p ON u.id = p.user_id  
7 WHERE p.id IS NULL;
```

The result grid displays the following data:

	id	username
▶	5	Aniya_Hackett
	7	Kassandra_Homenick
	14	Jadyn81
	21	Rocio33
	24	Maxwell.Halvorson
	25	Tierra.Trantow
	34	Pearl7
	36	Ollie_Ledner37
	41	Mckenna17
	45	David.Osinski47

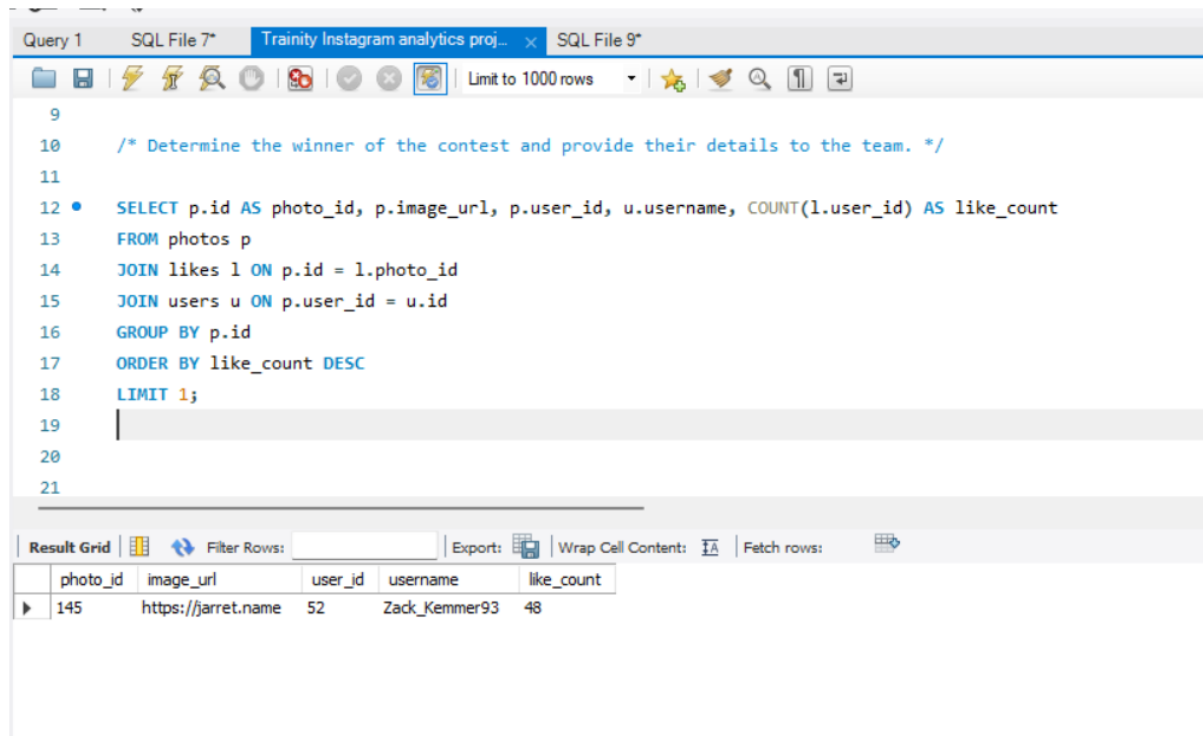
Insight:

These users have never posted. A personalized campaign could be launched to motivate them to become active contributors.

3. Contest Winner Declaration

The team has organized a contest where the user with the most likes on a single photo wins.

Task: Determine the winner of the contest and provide their details to the team.



The screenshot shows a SQL IDE window with a query editor and a results grid. The query is designed to find the user with the most likes on a single photo.

```
9
10  /* Determine the winner of the contest and provide their details to the team. */
11
12  • SELECT p.id AS photo_id, p.image_url, p.user_id, u.username, COUNT(l.user_id) AS like_count
13  FROM photos p
14  JOIN likes l ON p.id = l.photo_id
15  JOIN users u ON p.user_id = u.id
16  GROUP BY p.id
17  ORDER BY like_count DESC
18  LIMIT 1;
19
20
21
```

The results grid shows the following data:

photo_id	image_url	user_id	username	like_count
145	https://jarret.name	52	Zack_Kemmer93	48

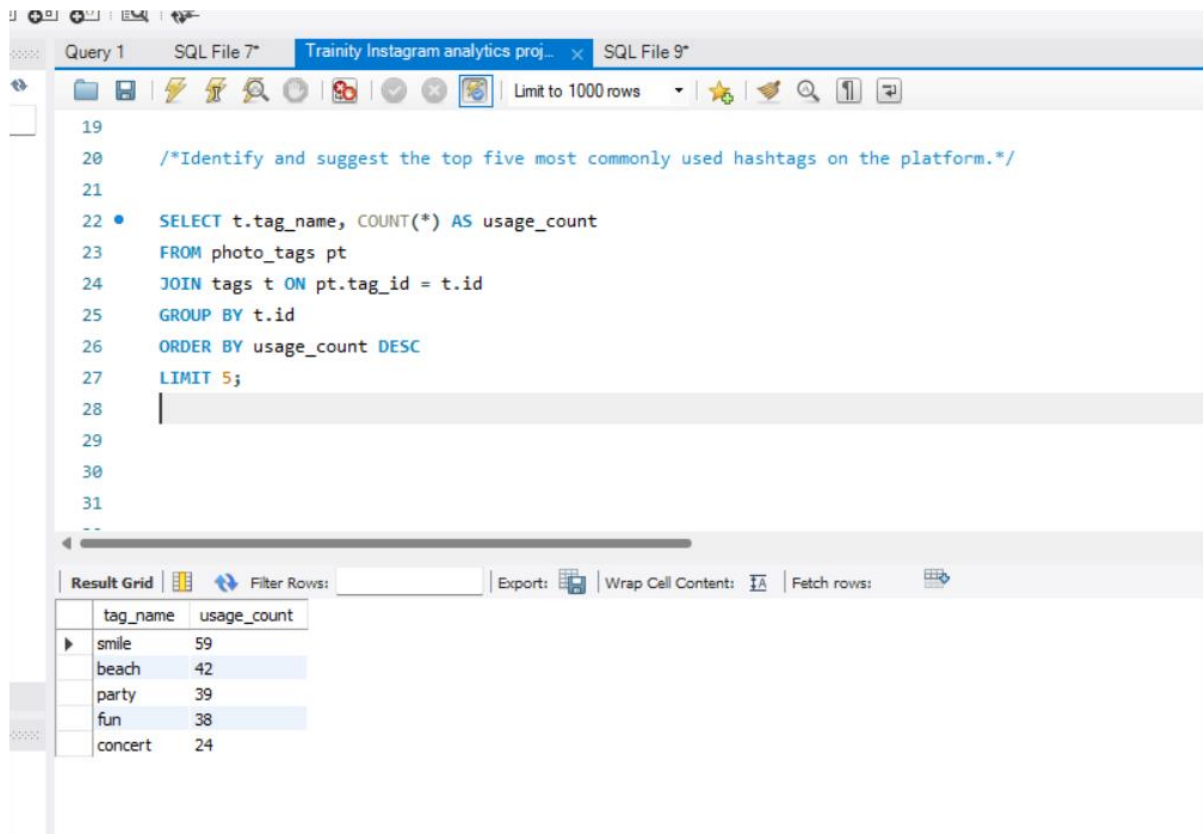
Insight:

This identifies the user with the most liked photo. We can declare contest winner to this user .

4.Hashtag Research:

A partner brand wants to know the most popular hashtags to use in their posts to reach the most people.

Task: Identify and suggest the top five most commonly used hashtags on the platform.



The screenshot shows a SQL IDE window with a query editor and a result grid. The query is designed to identify the top five most commonly used hashtags by joining the 'photo_tags' and 'tags' tables, grouping by tag ID, and ordering by usage count in descending order, limited to five results.

```
19
20  /*Identify and suggest the top five most commonly used hashtags on the platform.*/
21
22  • SELECT t.tag_name, COUNT(*) AS usage_count
23  FROM photo_tags pt
24  JOIN tags t ON pt.tag_id = t.id
25  GROUP BY t.id
26  ORDER BY usage_count DESC
27  LIMIT 5;
28
29
30
31
--
```

The result grid displays the following data:

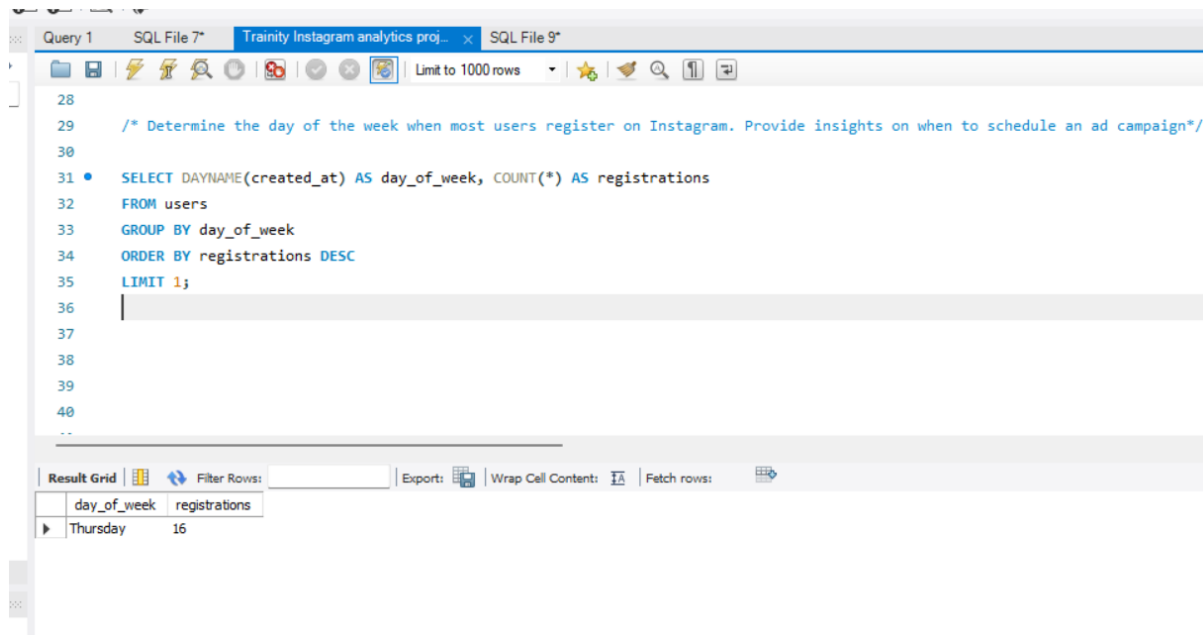
tag_name	usage_count
smile	59
beach	42
party	39
fun	38
concert	24

Insight: These hashtags are the most frequently used and are ideal users aiming for greater visibility.

5. Ad Campaign Launch:

The team wants to know the best day of the week to launch ads.

Task: Determine the day of the week when most users register on Instagram. Provide insights on when to schedule an ad campaign.



The screenshot shows a SQL IDE window with a query editor and a result grid. The query editor contains the following SQL code:

```
28
29  /* Determine the day of the week when most users register on Instagram. Provide insights on when to schedule an ad campaign*/
30
31  • SELECT DAYNAME(created_at) AS day_of_week, COUNT(*) AS registrations
32     FROM users
33     GROUP BY day_of_week
34     ORDER BY registrations DESC
35     LIMIT 1;
36
37
38
39
40
...
```

The result grid at the bottom shows the following data:

day_of_week	registrations
Thursday	16

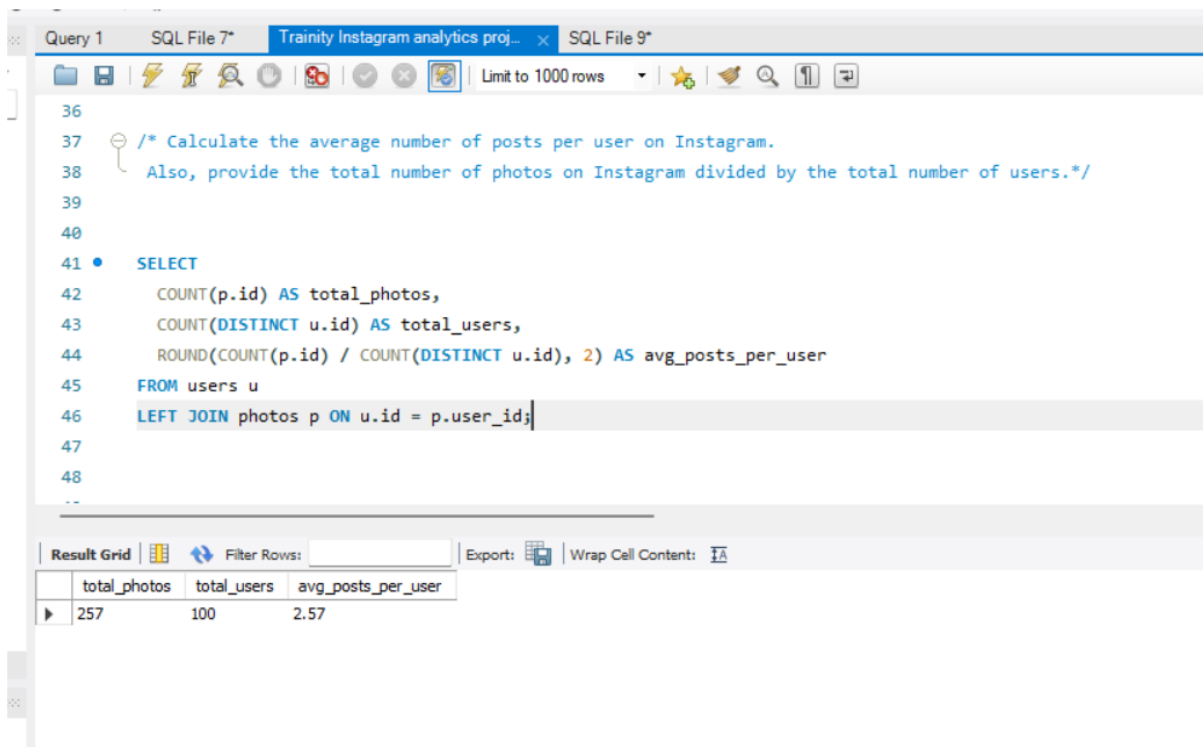
Insight: This result shows the day with the highest user registrations, which can guide the timing of marketing ads.

B) Investor Metrics:

1. User Engagement

Investors want to know if users are still active and posting on Instagram or if they are making fewer posts.

Task: Calculate the average number of posts per user on Instagram. Also, provide the total number of photos on Instagram divided by the total number of users.



The screenshot shows a SQL IDE window with a query editor and a result grid. The query is as follows:

```
36
37 /* Calculate the average number of posts per user on Instagram.
38    Also, provide the total number of photos on Instagram divided by the total number of users.*/
39
40
41 • SELECT
42     COUNT(p.id) AS total_photos,
43     COUNT(DISTINCT u.id) AS total_users,
44     ROUND(COUNT(p.id) / COUNT(DISTINCT u.id), 2) AS avg_posts_per_user
45 FROM users u
46 LEFT JOIN photos p ON u.id = p.user_id;
```

The result grid shows the following data:

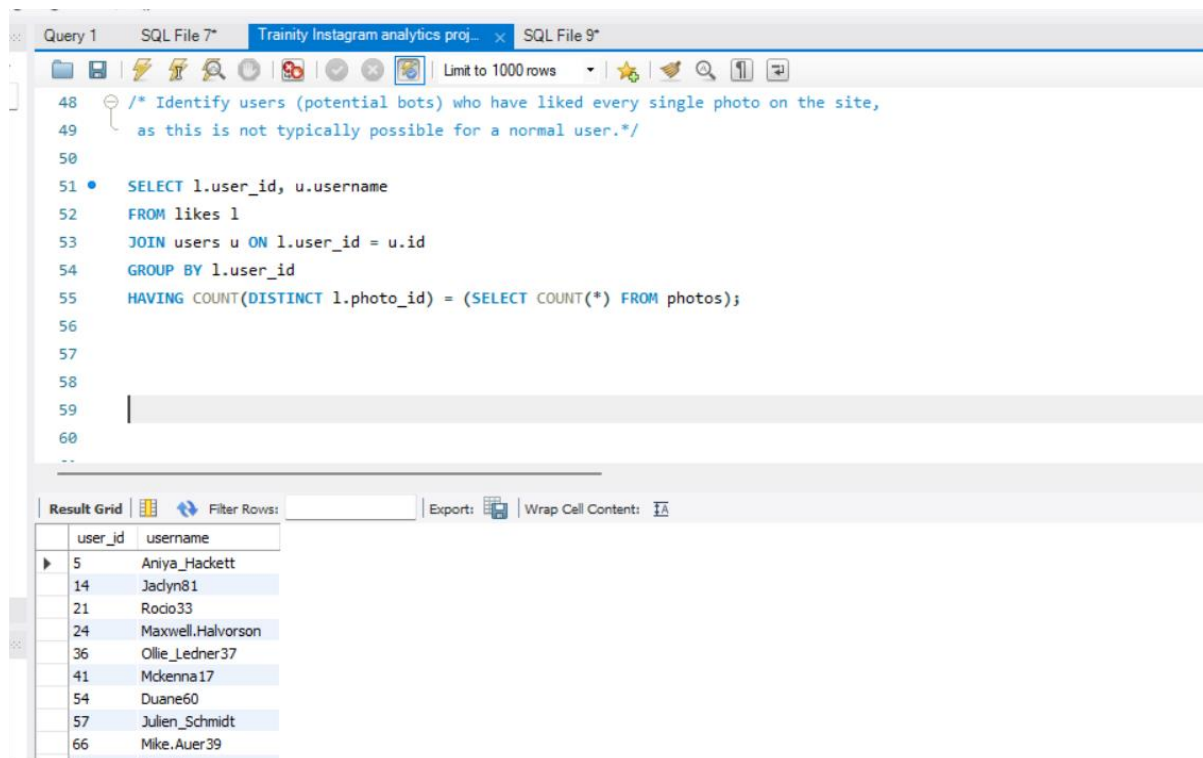
	total_photos	total_users	avg_posts_per_user
▶	257	100	2.57

Insight: This metric helps gauge user engagement across the platform.

2.Bots & Fake Accounts

Investors want to know if the platform is crowded with fake and dummy accounts.

Task: Identify users (potential bots) who have liked every single photo on the site, as this is not typically possible for a normal user.



The screenshot shows a SQL IDE window with a query editor and a result grid. The query is designed to identify users who have liked every single photo on the site, which is a task typically impossible for a normal user.

```
48 /* Identify users (potential bots) who have liked every single photo on the site,  
49 as this is not typically possible for a normal user.*/  
50  
51 • SELECT l.user_id, u.username  
52 FROM likes l  
53 JOIN users u ON l.user_id = u.id  
54 GROUP BY l.user_id  
55 HAVING COUNT(DISTINCT l.photo_id) = (SELECT COUNT(*) FROM photos);  
56  
57  
58  
59  
60
```

The result grid displays the following data:

user_id	username
5	Aniya_Hackett
14	Jadyn81
21	Rocio33
24	Maxwell.Halvorson
36	Ollie_Ledner37
41	Mckenna17
54	Duane60
57	Julien_Schmidt
66	Mike.Auer39

Insight: Users who have liked every photo may be bots or unusually active accounts. These should be reviewed further.

Result:

Working on this project helped me improve my SQL skills and gain deeper insights into how data analysis can support business goals. I learned how to structure efficient queries, interpret output meaningfully, and communicate findings in a clear way. It also gave me a practical perspective on analysing user behaviour.